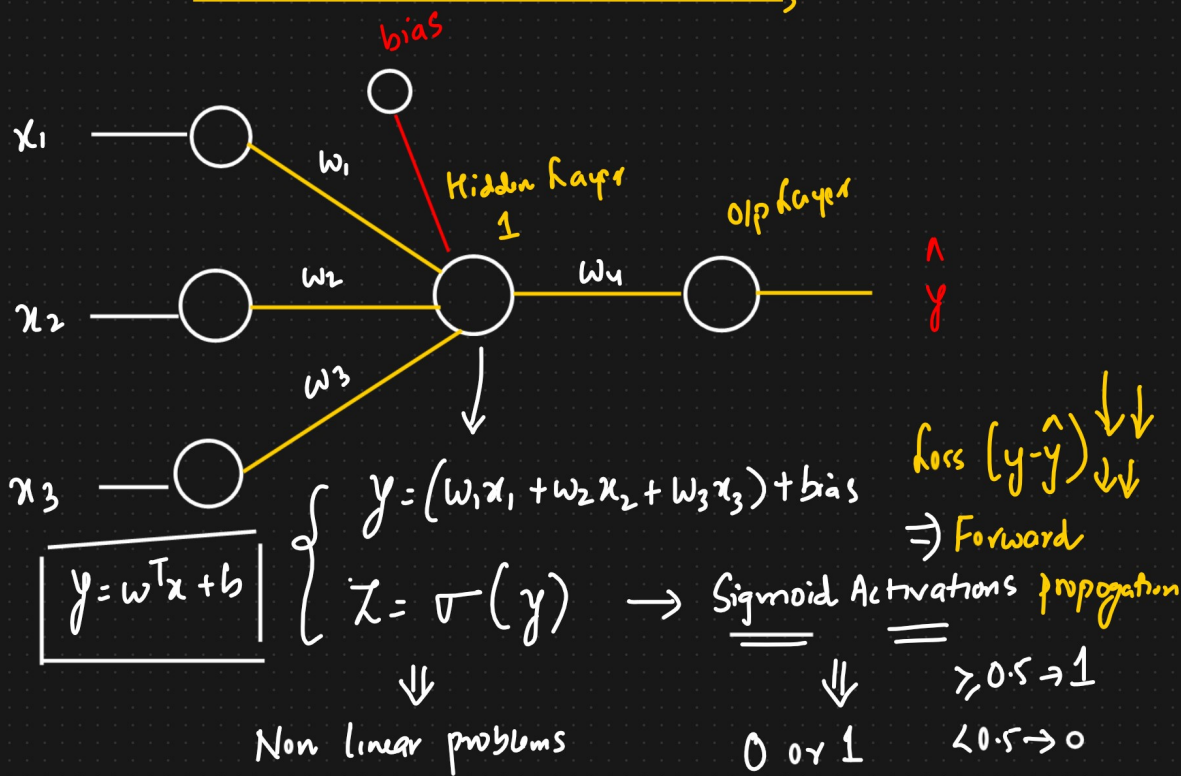


Day 2 - Deep Learning.

Agenda

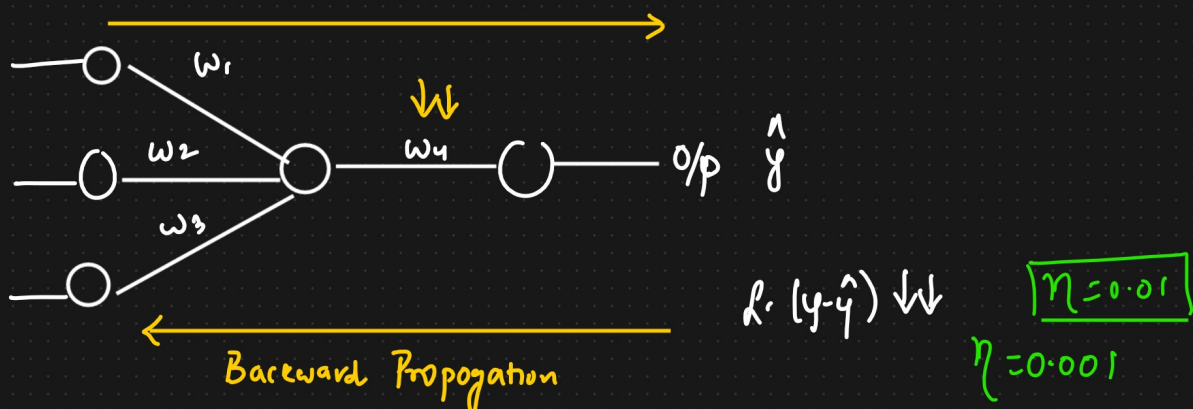
- ① Forward Propagation ✓
- ② Chain Rule of Derivative ✓
- ③ Vanishing Gradient Problem ✓
- ④ Loss functions ✓

Activation functions



② Backpropagation

- ① Weight updation formula
- ② Chain Rule of Differentiation ✓



① Weight updation formula

$$W_{\text{new}} = W_{\text{old}} - \eta \left[\frac{\partial h}{\partial W_{\text{old}}} \right]$$

Learning Rate

$\frac{\partial h}{\partial W_{\text{old}}}$ $\rightarrow$ slope

$\boxed{-ve \text{ slope}}$

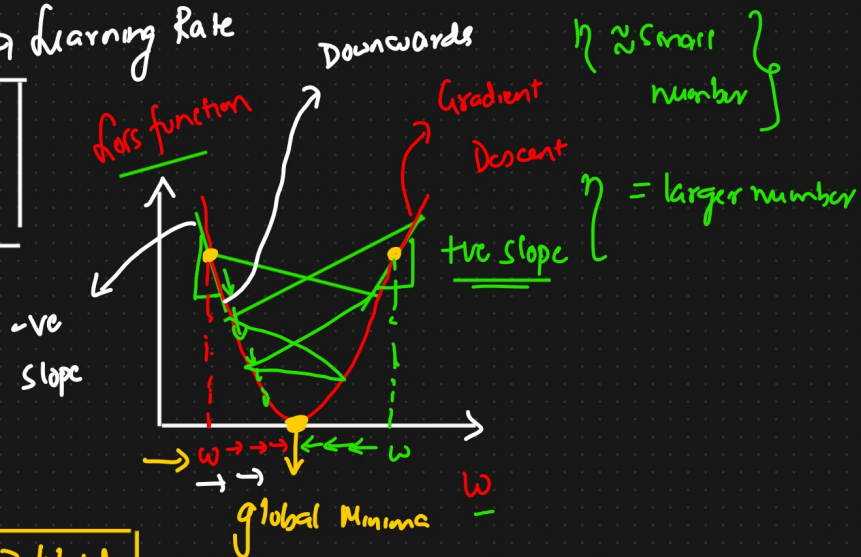
$$W_{\text{new}} = W_{\text{old}} - \eta (-ve)$$

$$= W_{\text{old}} + \eta (+ve)$$

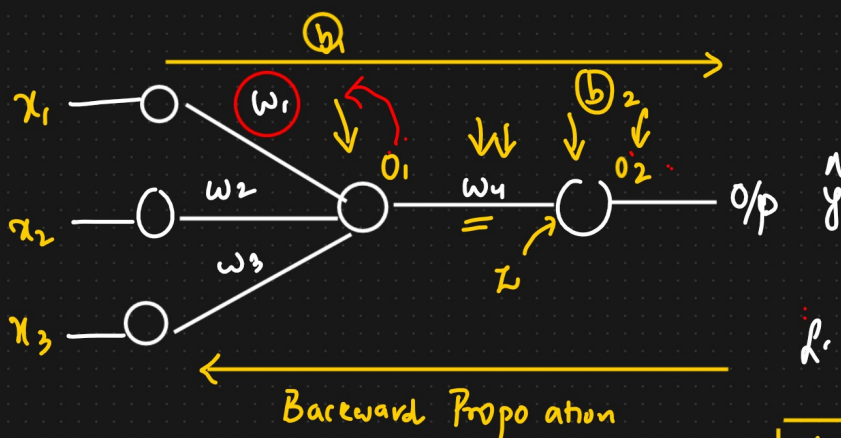
$$\boxed{W_{\text{new}} > W_{\text{old}}}$$

$$W_{\text{new}} = W_{\text{old}} - \eta (+ve)$$

$$\boxed{W_{\text{new}} < W_{\text{old}}}$$



② Chain Rule of Differentiation



$$W_{\text{new}} = W_{\text{old}} - \eta \left[\frac{\partial h}{\partial W_{\text{old}}} \right]$$

$$Z = \sigma(o_1 w_4 + b)$$

$$L(y, \hat{y})$$

$$W_{4 \text{ new}} = W_{4 \text{ old}} - \eta \frac{\partial h}{\partial W_{4 \text{ old}}}$$

$$b_{2 \text{ new}} = b_{2 \text{ old}} - \eta \frac{\partial L}{\partial b_{2 \text{ old}}}$$

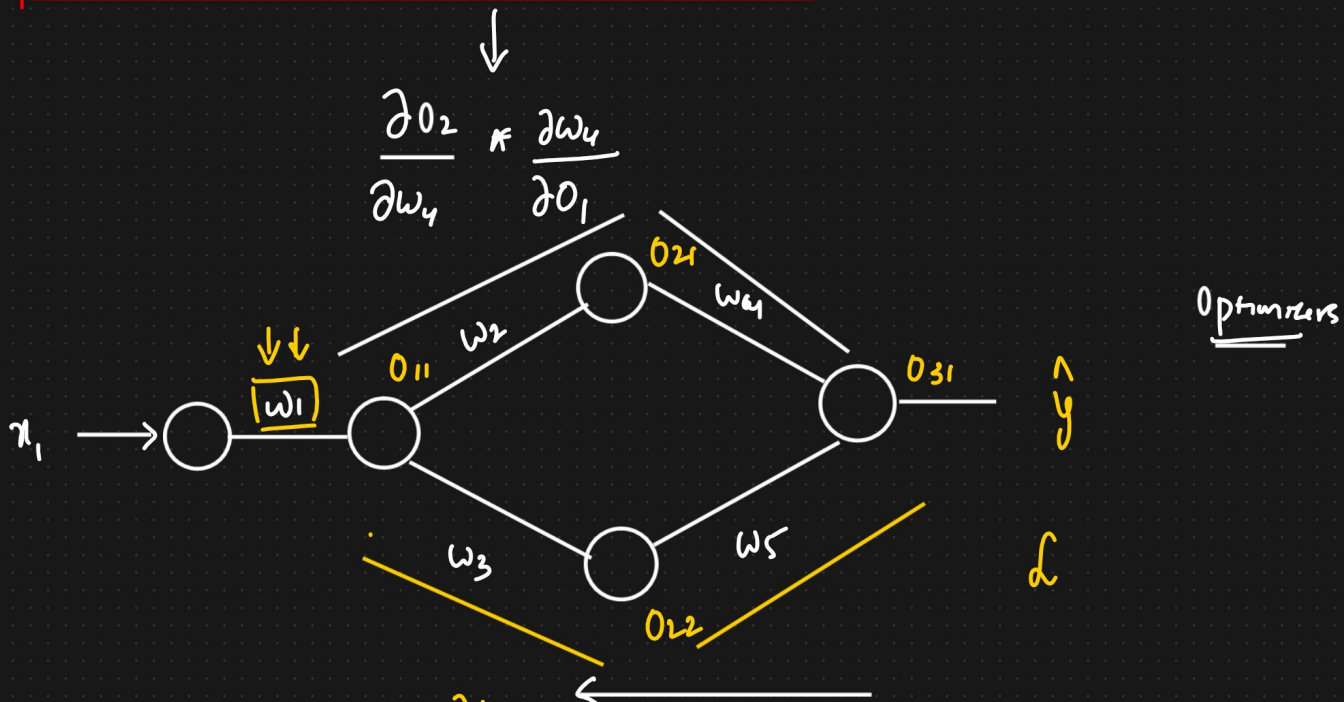
{ Chain Rule of Derivative }

$$\frac{\partial h}{\partial W_{4 \text{ old}}} = \frac{\partial h}{\partial o_2} * \frac{\partial o_2}{\partial W_{4 \text{ old}}}$$

$$w_{\text{new}} = w_{\text{old}} - \eta \left[\frac{\partial h}{\partial w_{\text{old}}} \right]_{\text{loss}}$$

$$\omega_{2_{\text{new}}} = \omega_{2_{\text{old}}} - \eta \frac{\partial L}{\partial \omega_{2_{\text{old}}}}$$

$$\frac{\partial \text{Loss}}{\partial w_{1, \text{old}}} = \frac{\partial \text{Loss}}{\partial z_2} * \frac{\partial z_2}{\partial w_{1, \text{old}}} = \frac{\partial \text{Loss}}{\partial w_{1, \text{old}}}$$



$$w_{i, \text{new}} = w_{i, \text{old}} - \eta \left[\frac{\partial L}{\partial w_{i, \text{old}}} \right]$$

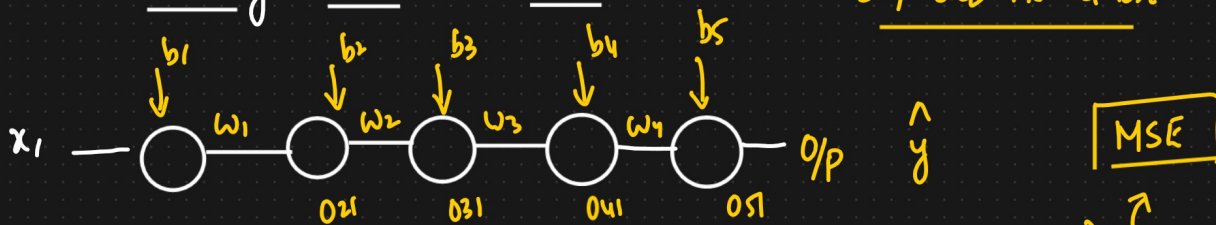
⇒ Chain Rule of Derivatives

$$\frac{\partial L}{\partial w_{old}} = \left[\frac{\partial L}{\partial z_1} * \frac{\partial z_1}{\partial z_2} * \frac{\partial z_2}{\partial z_3} * \frac{\partial z_3}{\partial w_{old}} \right]$$

$$+ \left[\frac{\partial L}{\partial \phi_1} * \frac{\partial \phi_1}{\partial \phi_2} * \frac{\partial \phi_2}{\partial \phi_3} * \frac{\partial \phi_3}{\partial \omega_{1,d}} \right]$$

③ Vanishing Gradient Problem

Sigmoid Activation



$$w_{\text{new}} = w_{\text{old}} - \eta \frac{\partial h}{\partial w_{\text{new}}}$$

$$\text{loss} = \frac{1}{2} (y - \hat{y})^2$$

$$o_{51} = \sigma \left[(o_{41} * w_4) + b \right]$$

↓ Sigmoid Activation

0 to 0.25

$$\frac{\partial h}{\partial w_{\text{new}}} = \frac{\partial h}{\partial o_{51}} * \frac{\partial o_{51}}{\partial o_{41}} * \frac{\partial o_{41}}{\partial o_{31}} * \frac{\partial o_{31}}{\partial o_{21}} * \frac{\partial o_{21}}{\partial w_1}$$

Derivative.

$$= 0.25 * 0.15 * 0.10 * 0.05 * 0.02$$

$$= \text{Small value}$$

$y = w^T x + b$

$$0 \leq \sigma(y) \leq 0.25$$



$$z = \frac{1}{1 + e^{-x}}$$

$$\rightarrow \begin{matrix} 0 \text{ or } 1 \\ \geq 0.5 & 1 \\ < 0.5 & 0 \end{matrix}$$

$$w_{\text{new}} = w_{\text{old}} - \eta (\text{small number})$$

Another Activation function.

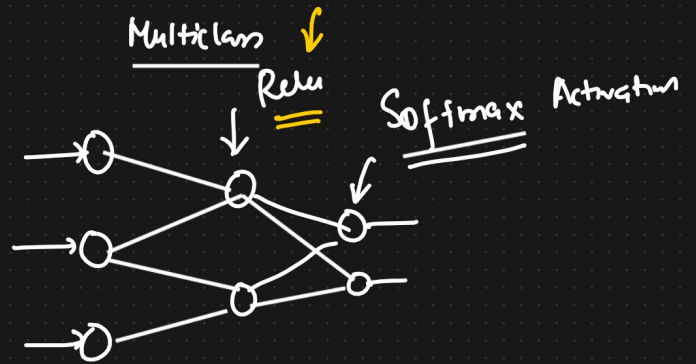
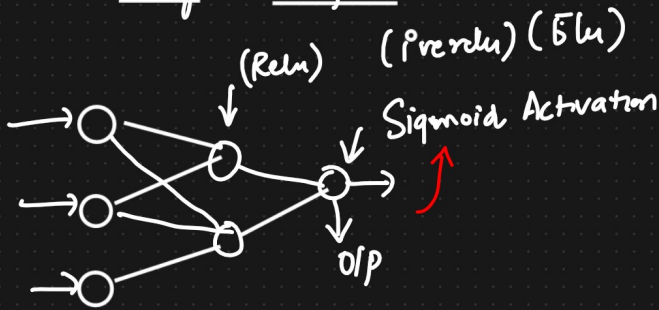
$$w_{\text{new}} \approx w_{\text{old}} \Rightarrow \text{Vanishing Gradient Problem}$$

↓
No change in weights

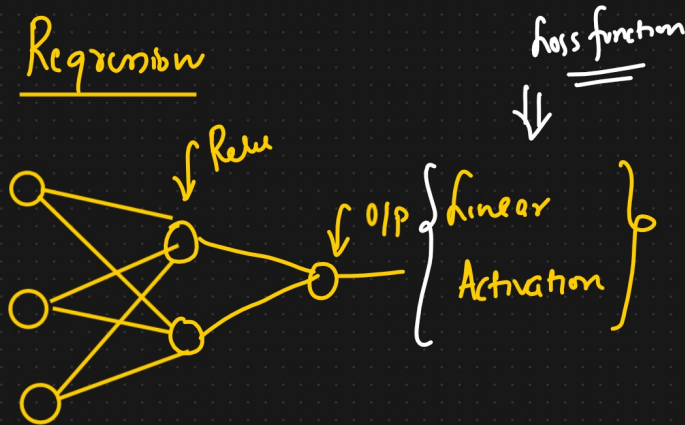
- ① Sigmoid
- ② Tanh
- ③ ReLU
- ④ Leaky ReLU
- ⑤ PReLU

Technique Which Activation fn we should Use

Binary classification



Regression



④ Loss functions

Deep Learning (ANN)

Regression			Classification		
<u>Dataset</u>			<u>Loss functions</u>		
Exp	Degree	Salary	Play	Study	Pass/Fail
10	phd	—	10	2	Fail
—	—	—	4	3	Fail
—	—	—	5	5	Maybe
			2	7	Pass

① Regression

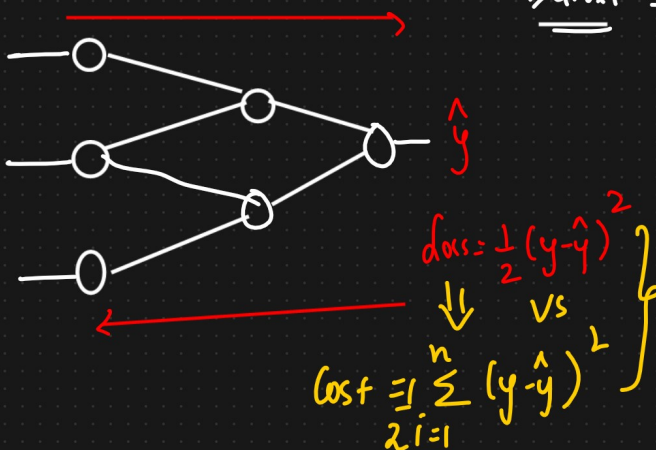
- ① MSE {Mean Squared Error}
- ② MAE {Mean Absolute Error}
- ③ Huber Loss

Loss function AND

COST Function

10 records
↓

Dataset = 100 records

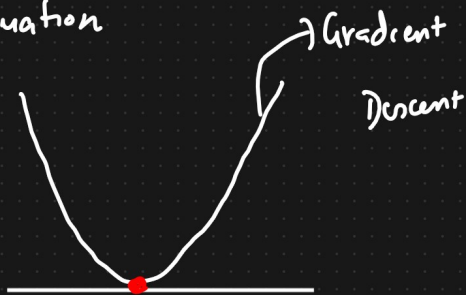


① Mean Squared Error (MSE)

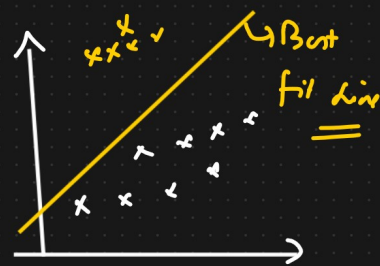
Loss function = $\frac{1}{2} (y - \hat{y})^2$
 ↓ Error
 Quadratic Equation

Cost function = $\frac{1}{2} \sum_{i=1}^n (y - \hat{y})^2$

$(a-b)^2 = a^2 - 2ab + b^2$
 ↓
 $ax^2 + bx + c$



Penalizing Error



Advantages

- ① Differentiable
- ② It has only 1 local or Global Minima.
- ③ It converges faster

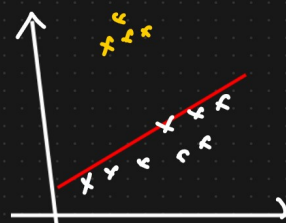
Disadvantage

- ① Not Robust to outliers

② Mean Absolute Error

loss fn = $\frac{1}{2} (y - \hat{y})$

Cost fn = $\frac{1}{2} \sum_{i=1}^n |y - \hat{y}|$



① Robust to outliers

$\Rightarrow$ Time consuming $\uparrow\uparrow$



$\Rightarrow$ Subgradient

③ Huber loss

① MSE

② MAE

outlier are not present

Hyperparameter

$$\text{loss} = \begin{cases} \frac{1}{2} (y - \hat{y})^2 & \text{if } |y - \hat{y}| \leq \delta \\ \delta |y - \hat{y}| - \frac{1}{2} \delta^2, & \text{otherwise} \end{cases}$$

③ Classification

CROSS ENTROPY $\begin{cases} \rightarrow \text{Binary Cross Entropy} \rightarrow \text{Binary classification} \\ \rightarrow \text{Categorical Cross Entropy} \rightarrow \text{Multiclass classification} \end{cases}$

① Binary Cross Entropy

$$\text{loss} = -y * \log(\hat{y}) - (1-y) * \log(1-\hat{y}) \Rightarrow \text{Logistic Regression}$$

$$\text{loss} = \begin{cases} -\log(1-\hat{y}) & \text{if } y=0 \\ -\log(\hat{y}) & \text{if } y=1 \end{cases} \text{ Binary classification}$$

$$\hat{y} = \frac{1}{1 + e^{-x}}$$

② Categorical Cross Entropy {Multi-class Classification Problem}

f_1	f_2	f_3	O/P	Good	Bad	Neutral	i = Row j = Column
2	3	4	Good	1	0	0	
5	6	7	Bad	0	1	0	
8	9	10	Neutral	0	0	1	

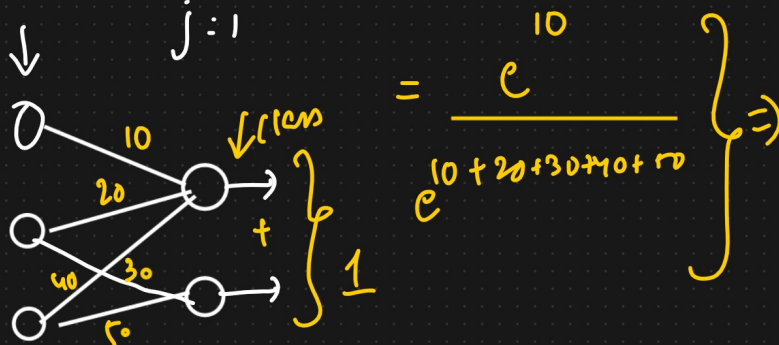
$$L(x_i, y_i) = - \sum_{j=1}^c [y_{ij}] * \ln(\hat{y}_{ij})$$

$$y_i = [y_{i1}, y_{i2}, y_{i3} \dots y_{ic}]$$

$$y_{ij} = \begin{cases} 1 & \text{if the element is in class.} \\ 0 & \text{otherwise} \end{cases}$$

$$\hat{y}_{ij} = \text{Softmax Activation} \rightarrow \text{O/p Layer}$$

$$\sigma(z) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}} \Rightarrow \text{Softmax Activation}$$



0.4, 0.5, 0.6

Conclusions

Relu, Softmax $\Rightarrow$ Multiclass } $\rightarrow$ Categorical Cross Entropy
Relu, Sigmoid $\Rightarrow$ Binary } $\rightarrow$ Binary Cross Entropy.

Linear Regression

Relu, Linear Activa $\rightarrow$ MSE, MAE, Huber Loss